

BUSN 43800 — Group Assignment 1, Part B: Specify the Problem

Team: [Team Name]

Project: DataVoy (旅数参考) — Consolidating Fragmented Official Tourism Data

Stakeholder

China-based tourism industry analysts and planners — e.g., market researchers at travel agencies, hotel groups, destination management organizations, and investment analysts covering travel and hospitality. These practitioners need timely, comparable holiday travel statistics to support pricing, marketing, and investment decisions.

Current state

To prepare a single holiday market briefing, an analyst must visit multiple official sources:

- Ministry of Culture and Tourism (文旅部): domestic travel volume and spending during holidays.
- Ministry of Transport (交通部): cross-region passenger flow by rail, road, waterway, and aviation.
- National Immigration Administration (移民局): entry/exit volumes, including visa-free foreigners.

Each source publishes in different formats (website articles, WeChat posts, PDFs), uses different units (亿人次 vs. 万人次), and defines similar concepts differently (e.g., “domestic tourists” vs. “cross-region travelers”). A typical analyst spends 2–4 hours per holiday copying numbers into Excel, converting units, aligning dates, and verifying sources.

Desired state

The analyst opens a single platform, selects a holiday and year, and immediately sees:

1. A unified, comparable table of tourism-related metrics across ministries.
2. Each number linked directly to its original official source.
3. Transparent notes on unit conversions and comparability.
4. Filters and export options for reports.

A single holiday briefing should take under 15 minutes instead of hours.

Constraints

- Source fragmentation: Data lives on separate ministry websites and social media with no unified API.
- Format inconsistency: Tables, paragraphs, and WeChat articles require different extraction approaches.
- Unit and definition mismatch: Metrics that sound similar measure different populations or time windows.
- Trust requirement: Users will not cite numbers unless they can verify the original government source.
- Update unpredictability: Release dates and data granularity vary by holiday and by ministry.

Gap

The gap is both time and reliability. Time to prepare a multi-source holiday briefing is currently 2–4 hours; it should be under 15 minutes. Manual copying and unit conversion also introduce avoidable errors. For an analyst covering 6–8 major holidays per year, this is roughly 15–30 hours of mechanical work that could be redirected to interpretation and decision-making.

Relevance

The gap matters on three dimensions:

- To the analyst: Hours of tedious collection work and stress over transcription accuracy.
- To the firm: Slower market response on pricing, staffing, and marketing campaigns.
- To decision quality: Errors in cross-source comparisons can lead to misjudged demand and misallocated resources.

Because holiday travel data directly affects revenue decisions in travel, hospitality, and transportation, faster and cleaner access is decision-relevant.

Persistence

The gap will not resolve itself. Data fragmentation is structural — different ministries manage different systems and have no incentive to build a shared tourism dashboard. The same comparison is needed every Spring Festival, May Day, and National Day Golden Week. No existing substitute combines direct source links, normalized units, and cross-ministry coverage for free.

Analytic objective

Element	Statement
Decision	Which holiday-period tourism trends should the analyst prioritize for pricing, marketing, or investment action?
Metric	Time to complete a sourced, comparable holiday briefing (hours → minutes); error rate in cross-source comparisons.
Population	China-based tourism analysts and planners tracking official holiday statistics across multiple ministries.
Time horizon	Within 24 hours of an official data release.
Action	The analyst exports a normalized table or chart with source links and uses it in an internal report.

Theory A — The gap is a search-and-aggregation problem

The discrepancy exists because the data is physically scattered. Analysts know what they need but waste time visiting multiple sites. A unified aggregator that collects and displays the numbers in one place will close the gap by reducing search and copy-paste effort. If Theory A is correct, DataVoy’s main value is coverage and convenience.

Theory B — The gap is a normalization-and-trust problem

A different account: analysts can already find the raw numbers; what they cannot do quickly is compare them confidently across ministries. Units differ, definitions differ, and release methodologies differ. If Theory B is true, merely listing numbers from multiple sources is insufficient — the product’s value comes from transparent normalization, clear source attribution, and explicit notes on comparability.

Distinguishing evidence

Test which theory dominates with a small user study:

- Condition 1: Users see raw, unnormalized numbers scraped from each ministry side by side.
- Condition 2: Users see normalized, sourced, unified data with unit conversions and comparability notes.

Measure task completion time, self-reported confidence, and willingness to cite the output in a work report. If Condition 2 substantially outperforms Condition 1 on confidence and citation willingness, Theory B is the stronger force.

Success criteria

1. Speed: Median time to assemble a sourced holiday briefing drops from 2–4 hours to ≤ 15 minutes.

- 2. Traceability: 100% of displayed metrics link directly to the original official source.
- 3. Accuracy: Zero unit-conversion or alignment errors in the displayed tables.
- 4. Usability: ≥70% of test users correctly answer a cross-ministry comparison question using DataVoy faster and more accurately than using official sources alone.

Feasibility

What evidence can we obtain in five days?

	Substitute in five days	Cost
by real	5–10 classmates or contacts complete a timed task	Smaller sample, but enough to measure time and co
days and	Focus on 2–3 high-impact modules (e.g., Spring Festival and National Day Golden Week, 文旅部 + 交通部)	Less breadth, but proves the core value loop
aping from	Manual curation of historical data plus a lightweight static/demo frontend	Not scalable, but sufficient for a working demo
arison orts	Spot-check every number against its source link	Labor-intensive but feasible

We will build a minimum viable version of DataVoy with historical data for selected holidays, filtering by year and holiday, source links on every number, and a short data note explaining conversions.

Honor Code Pledge

"I pledge my honor that I have not violated the Honor Code in preparation of this group project."

Signatures:

- [Team Member 1]
- [Team Member 2]
- [Team Member 3]
- [Team Member 4]

AI-Use Disclosure

We used AI tools (including Claude) to help structure and refine this problem specification. All claims, numbers, and source attributions were reviewed by the team.